

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 30

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Correct answer only.</i> • C (1)	1
01.2	<i>1 mark for each point.</i> • Stratospheric ozone is natural and beneficial: absorbs UV, protecting living organisms (1) • Tropospheric ozone is a secondary pollutant (from NO _x photochemistry) that is toxic to plants and humans and is a greenhouse gas (1)	2
01.3	<i>1 mark for each point.</i> • HCFCs are less chemically stable / contain hydrogen so they are broken down in the troposphere (1) • So most do not persist long enough to reach the stratosphere where UV would release their chlorine (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 108-112; AQA spec 3.2.1.3.	
02.1	<i>Correct answer only.</i> • B (1)	1
02.2	<i>1 mark for each point.</i> • TBT is highly toxic to marine molluscs / crustaceans and an endocrine disruptor altering growth and reproduction of oysters / whelks even at very low concentrations (1) • It bioaccumulated in shellfish and could pass to humans eating them (1) • It was assumed dilution would make it harmless (illustrates the need for the precautionary principle); now prohibited and replaced by copper (1)	3

Question	Marking guidance	Marks
02.3	<i>1 mark for each point.</i> • In the low-oxygen conditions of the spoil heap iron is in its soluble reduced form and is leached into the river (1) • In the river it is oxidised to insoluble iron oxide (orange sediment); the oxidation uses dissolved oxygen so the river is deoxygenated, killing aerobic organisms (1) • Control: drainage water is passed over mesh screens where the iron oxidises and is deposited before the water reaches the river; the solid iron is removed periodically (1)	3
02.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Lead-acid car batteries (recycled) (1) • Lead flashing on roofs / windows (1) • Radiation shielding (1)	2
02.5	<i>1 mark.</i> • They inhibit enzyme function, especially the enzymes of the nervous system / nerve cells (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11.	
03.1	<i>1 mark for each point.</i> • The water was recharged during or before the last ice age (10 000-20 000 years ago) when the climate was wetter (1) • Current recharge is little or none so any abstraction reduces the stored volume (1) • The large volume means shortages take a long time to appear, but eventually the supply runs out; ancient water is often saline, causing soil salinisation (1)	3
03.2	<i>1 mark for each point.</i> • Two satellites whose orbits are affected by the force of gravity (1) • The mass of water in aquifers below changes gravity, so changes in the orbit / distance between the satellites indicate changes in groundwater mass (1) • Advantage: the whole Earth is surveyed every 30 days, whereas borehole water-table measurements are slow and hard to coordinate over large areas (1)	3
03.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Low-pressure irrigation pipes to reduce leaks (1) • Growing crops that need less water than wheat / maize (1) • Charging consumers by volume used (metering) (1) • Inter-basin transfer of water 450 km from the Yangtze (1)	3
03.4	<i>1 mark.</i> • Trees intercept rainfall and transpire large volumes, reducing soil moisture and the water percolating down to the aquifer (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 119-121; AQA spec 3.2.2.1-3.2.2.2.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> • Affluent people buy more consumer goods / trivia / disposable items and throw items away rather than repairing them (1) • Built-in obsolescence: products designed to have short lives / wear out / go out of fashion so customers upgrade (1) • Disposable products (razors, lighters, pens) that must be replaced (1) • Over-packaging to make products attractive; also increases wastes from manufacture and mineral extraction (1)	4
04.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Properties of the waste: toxicity, flammability, degradability, radioactivity (1) • Population density (collection costs) (1) • Land availability for landfill (1) • Availability of recycling technology / markets (1) • Legislation / regulatory framework (landfill tax) (1) • Public willingness to recycle / environmental awareness (1) • Processing costs / household income (1)	3
04.3	<i>1 mark for each point.</i> • Advantage: aerobic decomposition produces little methane; nutrients / organic matter returned to soil (circular economy); reduces landfill volume (1) • Disadvantage: needs separation / collection of clean organic waste; odour / pests / pathogens if poorly managed; low-value product; not suitable for mixed waste (1)	2
04.4	<i>1 mark.</i> • Many different materials are mixed together / contaminated so identification and separation are costly and the quality of recovered material is reduced (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-282; AQA spec 3.4.3.2.12.	
05.1	<i>Correct answer only.</i> • Its total annual discharge (1)	1
05.2	<i>1 mark for each point.</i> • Public supplies cannot match consumer demand / the water is often cleaner than rivers (1) • It also reduces urban flooding after heavy rain; surplus can be used for aquifer recharge (1)	2
05.3	<i>1 mark for each point.</i> • The reservoir is created in the estuary so no farmland / settlements need to be flooded (1) • But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5.	

Question	Marking guidance	Marks
06.1	<i>2 marks for each point (partial credit for a partly correct point).</i> • Temperature: most chemical / biological reactions are faster at higher temperature so degradable pollutants (eg sewage) break down faster; but faster decomposition may increase deoxygenation (2) • Light: provides the activation energy for chemical reactions; photodegradation of pesticides (eg neonicotinoids break down fastest in sunlight); light drives photochemical smog reactions producing secondary pollutants (2) • Oxygen: needed for aerobic bacterial decomposition of organic wastes / sewage; oxidation of sulfide ores produces sulfur dioxide / acid mine drainage; oxidation state affects the solubility of metals (2)	6
06.2	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Pollutant ions adsorb onto (the surface of) clay particles / organic matter in soil or sediment (1) • So they are immobilised and cannot be absorbed by organisms / leached into water; radiation is absorbed in the ground (1) • But they may be released later if conditions change (eg storms disturbing sediment, low pH) (1)	2
06.3	<i>1 mark for each point.</i> • Many heavy metals (eg lead, aluminium) are more soluble at low pH (1) • So they are mobilised / leached from soil or sediments and can be absorbed by organisms / enter water bodies (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 240-242, 255; AQA spec 3.4.2.2.	
07.1	<i>1 mark for each point.</i> • About 40-50 traps (1) • Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable (1)	2
07.2	<i>1 mark for each point.</i> • Widely scattered results (large standard deviation) mean a single sample may not be representative (1) • More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed (1)	2
07.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Same type and size of pitfall trap, set for the same length of time (1) • Traps emptied at the same time of day / same weather conditions in both habitats (1) • Same number of traps and same method of positioning (random or systematic) in each habitat (1) • Other variables that could affect results (eg rainfall, temperature) monitored or controlled (1)	3
07.4	<i>1 mark for each point.</i> • Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally (1) • Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends (1) • The interval between samples should match the rate at which the factor changes, found from a preliminary study (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	

Question	Marking guidance	Marks
08.1	<i>1 mark for each point.</i> - Negative feedback: a change causes other changes that reduce the rate or size of the original change, re-establishing equilibrium (1) - Positive feedback: a change causes other changes that increase the original change (1)	2
08.2	<i>2 marks for each point (partial credit for a partly correct point).</i> - Increased low-level cloud: higher temperatures increase evaporation and condensation, producing more cloud; clouds have a high albedo so more sunlight is reflected and warming is reduced (2) - Increased photosynthesis: higher temperatures and CO₂ increase photosynthesis, removing CO₂; if stored in woody tissue atmospheric CO₂ rises less (2)	4
08.3	<i>1 mark for each point.</i> - Permafrost: frozen waterlogged Arctic soils contain dead organic matter that decomposed anaerobically producing methane trapped in the frozen ground (1) - Warming thaws the permafrost, releasing methane (a powerful greenhouse gas) and allowing further decomposition, causing more warming (1) - Methane hydrate: methane from decomposed organic matter in deep sea sediments is held as a solid under high pressure and low temperature (1) - If deep water warms the hydrate melts, releasing methane into the atmosphere and causing further warming (1)	4
08.4	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Reduced albedo as snow / ice melts (1) - Ocean acidification reducing coral survival and carbon sequestration (1) - Increased forest and peat fires releasing CO₂ (1) - Faster decomposition of soil organic matter releasing CO₂ (1) - Increased water vapour (a greenhouse gas) in warmer air (1) - Increased cirrus (high) cloud which traps infrared (1)	3
08.5	<i>1 mark for each point.</i> - A point at which natural processes triggered by human-caused warming become self-sustaining / unstoppable (1) - So that stopping the original human activity would no longer stop climate change (eg runaway permafrost melt, methane hydrate release, forest fires) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 103-104, 371-372; AQA spec 3.2.1.2, 3.6.1.	
09.1	<i>1 mark for each point.</i> - The underlying granite contains uranium whose decay produces the radioactive gas radon (an alpha emitter) (1) - Radon travels up cracks in the rock and through foundations into buildings (1) - It becomes concentrated inside enclosed / poorly ventilated buildings; control by ventilating foundations / sumps (1)	3
09.2	<i>1 mark for the correct answer.</i> $16 + 1.5 = 17.5\%$ (1)	1

Question	Marking guidance	Marks
09.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Where they live (geology / radon, altitude) (1) • Occupation (radiographer, nuclear worker, aircrew, miner) (1) • Medical history (number of X-rays / CT scans) (1) • Lifestyle / diet (eg shellfish near discharges); smoking (1)	2
09.4	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • X-rays carry a small risk of mutation / cancer but the benefit (diagnosis, avoiding invasive surgery) may outweigh it; risks can be reduced (shielding, dose limits) (1) • Symptoms of chronic low-dose exposure take a long time to develop (1) • The same symptoms (cancers) have other causes so effects are hard to attribute (1) • Accurate data on human exposure / doses is often unavailable (especially after accidents) (1) • People who benefit and people who take the risk may be different (eg uranium miners vs electricity consumers) (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 297-299; AQA spec 3.4.3.2.14.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Principles: dB is logarithmic ($10 \text{ dB} = 10x$; $3 \text{ dB} = 2x$) so small reductions in dB are large reductions in intensity; effects on humans (hearing, stress, behaviour) and wildlife (breeding birds, bats, cetaceans) • Aircraft design: high bypass ratio engines, chevron nozzles, hush kits, acoustic liners, fairings, composites, blended wing • Aircraft operation: take-off angle, flight paths avoiding urban areas, constant descent, reduced reverse thrust, night restrictions, Chapter noise limits, fines and noise charges, quota count • Airport design / location: away from population centres, taxi and engine-test areas away from housing, acoustic walls, baffle mounds, land-use restrictions, free acoustic insulation • Railway: track polishing, sound-absorbing ballast and suspension, pantograph fairings, wheel lubrication, composite brakes • Road: porous / rubber asphalt, acoustic fences, embankments, tree planting, aerodynamics, engine insulation, traffic management and routing, double glazing • Industrial: silencers on compressors, pressing instead of stamping, nylon rollers, drilling instead of pile-driving, bubble curtains, timing of blasting, acoustic mats / curtains, remote operation, ear protection • Domestic: acoustic absorbers in appliances, volume limiters, selection of quieter appliances, ear defenders • Evaluation: which approaches are most effective / cost-effective; conflicts (airports near cities for convenience); legislation and monitoring (Leq 57 dB contours) needed to enforce	9
10.2	<i>1 mark.</i> • The area within which the average (equivalent continuous) sound level between 07:00 and 23:00 in summer exceeds 57 dB - taken as the onset of significant community annoyance (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 286-295; AQA spec 3.4.3.2.13.	

Question	Marking guidance	Marks
11.1	Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • How properties predict behaviour: dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • Control based on properties: persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones; systemic pesticides and seed coatings • Heavy metals: lead uses discontinued, storage at high pH to reduce solubility; mercury chemical form; radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays); oil properties (floats, insoluble) exploited by booms and skimmers; density in cyclone separators • Principles: CPA and CGM based on properties; emission location / timing; ALARA and BATNEEC; polluter pays; precautionary principle • New technologies: satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • Evaluation: prevention better than cure; limits of remediation; cost-effectiveness; contrast success of ozone (Montreal) with climate change; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	25

Question	Marking guidance	Marks
11.2	Soil is often described as a non-renewable resource. Discuss the causes and consequences of soil degradation and evaluate the strategies that can be used to manage soils sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Features of fertile soil: water, nutrients, air, organic matter, pH, biota, texture, structure, depth; slow rate of soil formation vs rapid erosion (dynamic equilibrium broken) • Causes of degradation: vegetation removal, ploughing vulnerable soils, overgrazing, reduced organic matter and biota, cultivating steep slopes, compaction, salinisation from irrigation, nutrient depletion, pollution • Types of erosion: rain splash, runoff, wind, slumping; consequences: reduced productivity, river and reservoir sedimentation, flooding, coastal sedimentation, particulates, desertification, landslides • USLE as a tool for estimating loss and testing management • Management: long-term crops, low / zero tillage, contour ploughing, tied ridging, terracing, stone rows, windbreaks, multicropping / strip cropping, mulching, organic matter, livestock control, drainage / pH / nutrient management, conservation of soil biota • Links to sustainable agriculture, carbon storage, hydrological cycle, circular economy • Evaluation and judgement on which strategies are most effective in different environments Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169, 328-337; AQA spec 3.2.5, 3.5.1.3, 3.5.1.5.	25